Fama-French Model

MGMT 675: AI-Assisted Financial Analysis

RICE | BUSINESS
Jones Graduate School of Business

Outline

- Does the CAPM work?
- Fama-French 3-factor model
- Fama-French 5-factor model
- Momentum factor
- Cost of equity capital with Fama-French model

Does the CAPM work?

- learn-investments.rice-business.org pulls from French's data library (and other things)
- Industry betas do not match average returns
- Size and book-to-market sorted portfolios

What could be wrong?

- CAPM regression for stock i:

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- Fama-French (1993) said we don't know, but we do know that small stocks beat big stocks on average and high book-to-market (value) stocks beat low book-to-market (growth) stocks on average.
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- Use the small minus big (SMB) and high-minus-low (HML) returns as proxies for the unknown risk factors.

Other factors

- Fama-French (2015): slow-growing companies beat fast-growing companies (growth in terms of assets) and profitable companies beat unprofitable companies (ROE or similar).
  - CMA = conservative minus aggressive (slow-growing minus fast-growing)
  - RMW = robust minus weak (profitable minus unprofitable)
- Carhart (1997): momentum factor (past winners minus past losers) called UMD or MOM
  - Evidence on momentum returns

Implementing the Fama-French 5-factor model

- Let's not use momentum and stick with the Fama-French 5-factor model.
- Steps:
  - Estimate the factor risk premia -
    - get longest possible data history on monthly factor returns and compute means.
  - Estimate the factor exposures over ten-year window:
    - Get monthly stock prices from yfinance 0.2.54 and compute returns
    - Fix date formats and decimal/percentage and merge with factor returns
    - Filter to last ten years for which everything is available
    - Run multivariate regression of stock return on factors
  - Get the 3-month or 10-year Treasury yield
  - Compute

$$r_f + \beta_{\text{Mkt-RF}}\overline{\text{Mkt-RF}} + \beta_{\text{SMB}}\overline{\text{SMB}} + \beta_{\text{HML}}\overline{\text{HML}} + \beta_{\text{CMA}}\overline{\text{CMA}} + \beta_{\text{RMW}}\overline{\text{RMW}}$$

Is the Fama-French model used?

- Morningstar uses it to validate its quantitative ratings
- Institutional investors use it to evaluate fund managers (next class)
- But firms predominantly use the CAPM (see Corporate Finance and Reality)